

Laboratory Procedure 101

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Basic Laboratory practices

1 Good Laboratory protocols

- Be punctual and attentive during laboratory sessions.
- **Handle all instruments and apparatus with care.**
- Record observations neatly, with correct units and significant figures.
- Use scientific notation where appropriate for very large or very small numbers.
- **Always mention the scale** in the legend while plotting graphs.
- **Ensure safety protocols are followed at all times.**

2 Scientific Notation

In laboratory work, measured quantities often span a wide range of magnitudes. To express such values clearly and concisely, we use **scientific notation**. In this system, a number is written in the form:

$$N = a \times 10^n$$

where a is a real number such that $1 \leq |a| < 10$, and n is an integer representing the order of magnitude.

Advantages of Scientific Notation

- It provides a compact and standardized way of expressing very large or very small numbers.
- It makes comparison of orders of magnitude straightforward.
- It reduces errors in recording and interpreting experimental data.
- It simplifies calculations involving multiplication and division.

Rules for Usage

1. Always express the coefficient a with the appropriate number of **significant figures** reflecting the precision of the measurement. For example:

$$0.000\,345 \rightarrow 3.45 \times 10^{-4}$$

2. Use scientific notation consistently in data tables, graphs, and calculations.
3. Units must always be retained when expressing values. Example:

$$6.022 \times 10^{23} \text{ mol}^{-1}$$

4. When reporting results, match the power of ten in both numerator and denominator if possible, to minimize confusion.

Examples

- The radius of an electron: $2.82 \times 10^{-15} \text{ m}$
- Avogadro's number: $6.022 \times 10^{23} \text{ mol}^{-1}$
- Speed of light: $2.998 \times 10^8 \text{ m/s}$

3 Procedure to Plot a Good Graph

1. **Choose axes:** Put the independent variable on the x-axis and the dependent variable on the y-axis. Use SI units.
2. **Select a suitable scale:** Use simple, evenly spaced tick increments (1, 2, 5, 10, ...). Make full use of the graph area so the data spread occupies most of the plotting region.
3. **Label axes:** Always show the quantity and unit, e.g. V (V) or I (A).
4. **Plot points accurately:** Mark each data point clearly (dot or small cross). Avoid large symbols that obscure accuracy.
5. **Best-fit line or curve:** For linear relationships, draw a best-fit straight line (not a line joining the points). For nonlinear data, draw a smooth curve that follows the trend.
6. **Show uncertainties:** Where applicable, draw error bars representing measurement uncertainty.
7. **Extract quantities:** Determine slope, intercept, and uncertainties from the graph. Use $\text{slope} = \Delta y / \Delta x$ for linear fits.
8. **Neat presentation:** Use a ruler for axes and lines, include a title, and keep the graph free of unnecessary clutter.

Solar cell $I - V$ practice

The $I - V$ characteristics of solar panel is governed by

$$I = I_L - I_s \left[\exp \left(\frac{eV_L}{k_B T} \right) - 1 \right]$$

where I_L is the photo-current, V_L is the voltage across the load resistance and T is the operating temperature.

Measurements

The following set of measurements for voltage and current are recorded for a solar panel in Table. 1.

Voltage V (V)	Current I (mA)	Power $P = VI$ (mW)
0.63	4.0	0
1.05	3.6	3.6
...	...	...
2.85	0	0

Table 1: Measured voltage and current for solar panel. Also the power vs current is plotted.

Plot

The plot in Figure. 1 shows the data points and the best-fit line for each of $I - V$ and $P - V$.

1. The best fit line should be a smooth line with points on either side. It need **not** pass through all the points.
2. We need to maximize graph paper utilization so that the data points are widely separated and confusion is minimized. To achieve this, a **proper scale** should be chosen. Consider the 1 cm squares as bricks of building material and you need to build a home on the available plot. Draw each axis two bricks inside from the graph paper edge to avoid punching problems. After drawing the axes, count the available land area in units of bricks along X and Y. Let us say we have 21 bricks along X and 16 along Y.
 - (a) Along X, find the range of data to be plotted. Lets say for voltage, we have data range from 0 V to 2.85 V. We can approximate range to [0 V, 3 V]. So the brick size is $3/18 = 0.16$ V. For convenience, we can approximate it to 0.15 V.
 - (b) Along Y, find the range of data to be plotted. Lets say for current, we have data range from 0 V to 4 mA. So the brick size is $4/16 = 0.25$ mA.

Analysis

The analysis is shown in Figure. 2.

How to choose proper scale?

Aim is to maximize utilization of graph area.

Graph has 21 (1 cm) bricks along X and 16 along Y.

Quantity	Range	#Bricks	Brick-size
Voltage	[0 V, 2.85 V]	21	
	~[0 V, 3 V]	~18	$(3-0)/18 = 0.16 \text{ V}$ ~ 0.15 V
Current	[0 mA, 4 mA]	16	$(4-0)/16 = 0.25 \text{ mA}$
Power	[0 mW, 6.7 mW] ~[0 mW, 8 mW]	16	$(8-0)/16 = 0.5 \text{ mW}$

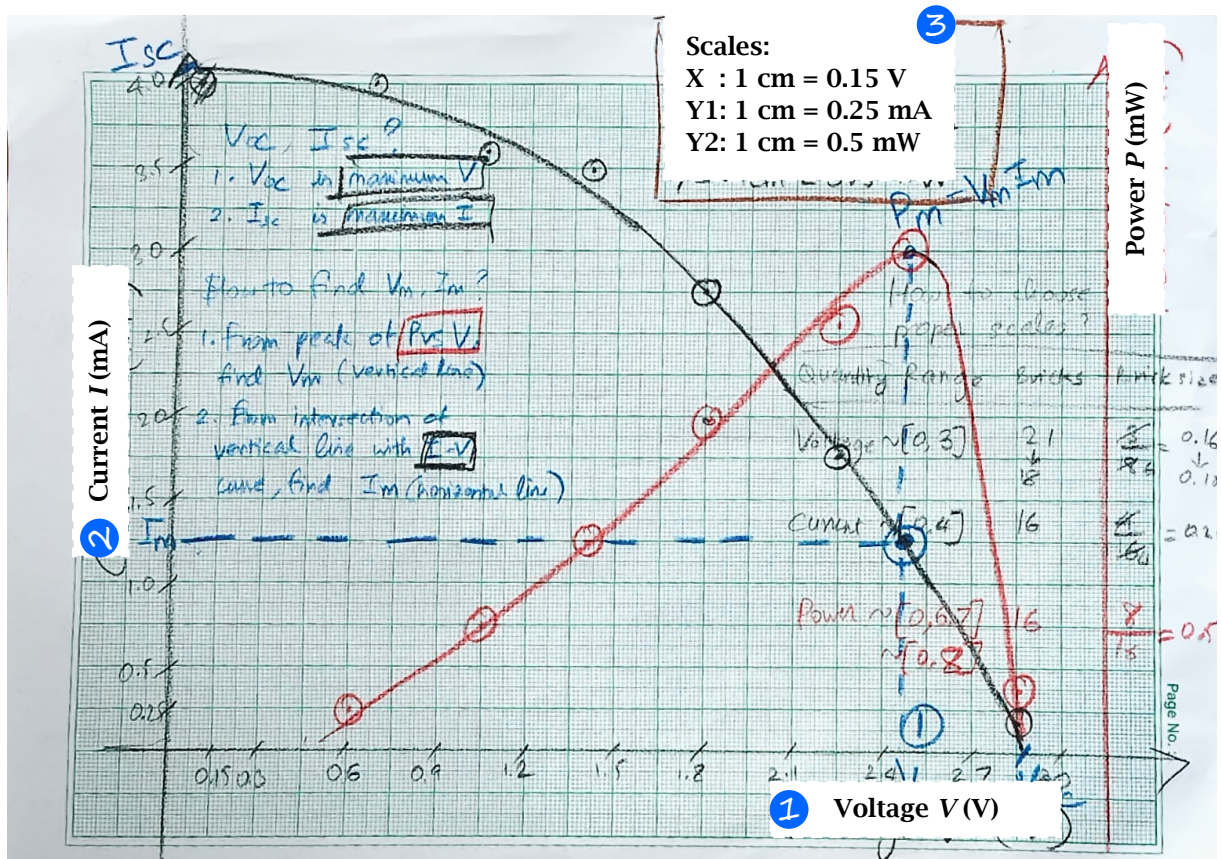


Figure 1: Solar cell practice.

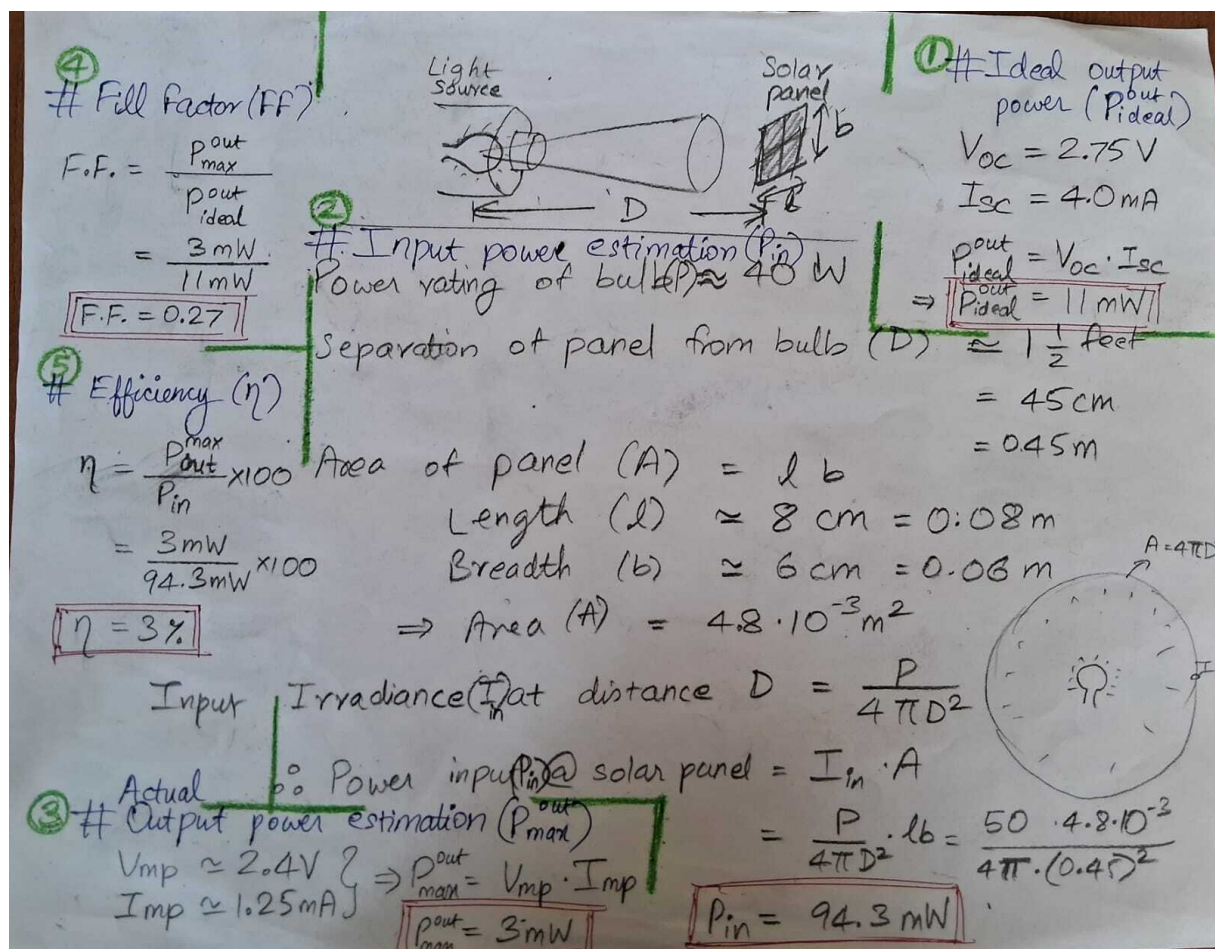


Figure 2: Solar cell practice analysis.

Advanced Laboratory protocols

4 Finding Constants of a Straight Line Using Error Analysis

When two physical quantities are related linearly, their relation can be expressed as:

$$y = mx + c$$

where

- m is the slope (constant of proportionality),
- c is the intercept on the y -axis,
- x and y are the measured variables.

4.1 Experimental Data

During the experiment, pairs of values (x_i, y_i) are recorded. Each measurement has an associated error:

- Δx_i : least count or uncertainty in measurement of x ,
- Δy_i : least count or uncertainty in measurement of y .

4.2 Best-Fit Straight Line

A graph of y vs. x is plotted. The slope m and intercept c can be determined using the **method of least squares**:

$$m = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2}$$
$$c = \frac{\sum y_i \sum x_i^2 - \sum x_i \sum x_i y_i}{N \sum x_i^2 - (\sum x_i)^2}$$

where N is the number of data points.

Expressions in Terms of Averages

Let

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$$

be the mean values of the data. Then, the slope m and intercept c are given by:

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad c = \bar{y} - m\bar{x}.$$

4.3 Error in Slope and Intercept

The deviations of observed points from the line are:

$$d_i = y_i - (mx_i + c)$$

The sum of squared deviations is used to estimate the uncertainty in slope and intercept.
The standard error of estimate is:

$$\sigma = \sqrt{\frac{\sum d_i^2}{N - 2}}$$

The error in slope is given by:

$$\Delta m = \sigma \sqrt{\frac{N}{N \sum x_i^2 - (\sum x_i)^2}}$$

The error in intercept is:

$$\Delta c = \sigma \sqrt{\frac{\sum x_i^2}{N \sum x_i^2 - (\sum x_i)^2}}$$

5 Significant Figures

In experimental physics, every measured quantity has an associated **uncertainty**. The number of **significant figures** used in reporting results must reflect this uncertainty.

1. **Definition:** Significant figures are the digits in a number that carry meaningful information about its precision. This includes all certain digits and the first uncertain digit.
2. **Rules for Use:**
 - All non-zero digits are significant.
 - Zeros between non-zero digits are significant.
 - Leading zeros are not significant; they only indicate the position of the decimal point.
 - Trailing zeros after a decimal point are significant.
3. **In Calculations:**
 - For multiplication or division: the result should have as many significant figures as the measured value with the *least number* of significant figures.
 - For addition or subtraction: the result should be rounded off to the *least precise decimal place* among the quantities involved.
4. **In Reporting Final Results:**
 - The uncertainty (error) should be quoted to *one significant figure* (occasionally two if the first digit is 1 or 2).
 - The measured value should be reported up to the same decimal place as the uncertainty.

Example: Counting Significant Figures

- 1234 $\rightarrow$ 4 significant figures (all non-zero digits).
- 0.00456 $\rightarrow$ 3 significant figures (leading zeros are not significant).
- 2.300 $\rightarrow$ 4 significant figures (trailing zeros after the decimal point are significant).
- 7.0890 $\rightarrow$ 5 significant figures (zeros between non-zero digits and the final zero are significant).
- 1.20×10^3 $\rightarrow$ 3 significant figures (scientific notation helps to clearly indicate them).

Example:

If resistance is found to be

$$R = 12.36 \, \Omega \pm 0.07 \, \Omega,$$

the correct way to report is

$$R = 12.36 \pm 0.07 \, \Omega$$

and not 12.360 ± 0.07 , since extra digits suggest unwarranted precision.

5.1 Final Result

The constants of the line are written with their uncertainties as:

$$m \pm \Delta m, \quad c \pm \Delta c$$

This ensures the linear relation is reported within the limits of experimental accuracy.

Note: This method is commonly applied in laboratory experiments such as verification of Ohm's law, Hooke's law, and determination of refractive index, where linear relations are analyzed.

Notes and Tips

- Always record data in scientific notation when the number is too large or too small for convenient decimal representation.
- Always include units and report final values with **appropriate significant figures** and uncertainty.